

Forward Deployed Engineer — Intern

Take-home case: Kestrel Health Insurance

The situation

Kestrel Health Insurance is a Blue Machines customer. We run their inbound policyholder support line — an AI voice agent handling four things: **claim status**, **network hospital lookup**, **renewal premium quotes** and **policy details**. Anything the agent cannot handle is transferred to a human agent on Kestrel's support floor.

The number that matters to Kestrel is **containment** — the share of calls the agent resolves without transferring to a human. Every transfer costs them money and adds to the queue.

On 31 August, Kestrel's Head of Customer Service escalated:

“Containment has fallen off a cliff over the last three weeks. Your bot has clearly got worse — it isn't understanding our customers any more. Our floor is drowning. We need this back where it was, and we need to know what you are going to do about it.”

You are the FDE on this account. You have three weeks of data. **Your job: work out what is actually happening, and decide what to do about it.**

What you have been given

File	What it is
calls.csv	Every call in the period — 3,544 rows. One row per call.
tool_calls.jsonl	Every backend tool invocation made by the agent, one JSON object per line. Joins to calls.csv on call_id.
customer_weekly_kpis.csv	Kestrel's own weekly summary — the report their Head of CS is looking at.
deployments.csv	Every change Blue Machines made to the deployment in the period.
support_tickets.csv	Tickets raised by Kestrel's operations and IT teams.
sample_transcripts.md	Six call transcripts chosen by Kestrel's ops team.

Column notes for **calls.csv**:

Column	Meaning
intent	What the caller wanted
claim_id	The claim reference given by the caller, where applicable
agent_version	Which version of our agent prompt handled the call
asr_confidence_avg	Average speech-recognition confidence across the call
avg_response_ms	Average time from caller finishing speaking to agent starting to speak
outcome	contained, transferred_human, or abandoned
transfer_reason	Set only where the call was transferred

In **tool_calls.jsonl**, the **response** field is the payload our agent received back from Kestrel's systems.

What to produce

A ten-minute presentation. Slides, a document, a notebook — whatever you would actually use. Bring it on a laptop.

Cover four things:

1 · What is happening	What is the actual cause of the containment drop? Show us how you know, not just what you concluded.
2 · What it is costing	Put a number on the impact.
3 · What you would do	Immediate, and after that. Be specific about what you would change and where.
4 · What you would tell Kestrel	Write the update you would send their Head of Customer Service. Keep it short.

Rules

- **Use any tool you like** — ChatGPT, Claude, Cursor, Copilot, whatever you normally use. We are not testing whether you can work without them.
- **Be ready to say what you used them for, and where you disagreed with them.** We will ask. A candidate who cannot explain their own analysis did not do it.
- **Spend about three to four hours on it.** We are not looking for polish. We are looking for whether you found the right thing and can defend it.
- **Nothing is submitted in advance.** We see your work for the first time when you present it. Do not email anything beforehand.
- You may find that not every piece of data is relevant. That is deliberate.
- If you think something important is missing from the pack, say so in the presentation — and say what you would have done with it.

After the presentation

We will spend the rest of the session on your approach and your reasoning: why you looked where you looked, what you ruled out and why, and what you would do if one of your assumptions turned out to be wrong. Expect to be pushed on your conclusions. That is the interview, not a test of whether you can be rattled.

Bring questions for us.

Kestrel Health Insurance is a fictional account created for this exercise. All data in the pack is synthetic. No Blue Machines customer data is included.